
Retention You Can Predict, Scope and Delete: a (μ, γ, T) Law and a Structural Deletion Guarantee for a Physics-Structured Memory Store

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Abstract

Existing memory systems enforce retention via explicit bookkeeping policies. Forgetting or deletion thereby relies on methods like decay factors, expiry timestamps, or learned forget gates, which dictate when a specific entry expires. However, these programmatic rules do not specify the physical rate of value degradation, the continuous dynamical effect of tuning retention parameters, or the exact residual state of the network following an item’s removal. In this work, we resolve these ambiguities by formalizing forgetting as a dynamical property of a physics-structured memory store: specifically, one produced by a damped symplectic step of a learned Hamiltonian. Our retention policy is defined by a computable physical budget parameterized by three variables: the stored direction’s spectral mass μ , the damping γ , and the temperature T .

We demonstrate three core properties of this architecture. First, the retention dial possesses a computable optimum: retention half-life is non-monotonic in friction, forming a V-curve minimized at $\gamma_{\text{crit}} = 2\varepsilon\mu$. This curve spans approximately 11 orders of magnitude in μ^2 and holds across both designed and emergent (learned-MLP) units. Second, retention is rigorously scopeable: at $T > 0$, friction preserves memory while temperature erases it. A localized friction hole thus acts as a memory vault, demonstrating a $107.77 \pm 4.78\times$ retention factor on designed architectures. Finally, removal is purely structural: under canonical Blelloch and Golovin [2007] placement, store-level deletion is exact. Post-deletion, the store is byte-identical to a state that never held the item, governed by defined operational conditions, on a designed, non-learned 3-dimensional datastore at capacities 8–64.

1 Introduction

To contextualize the operational constraints of modern memory systems, we begin with deployed retention policies. Most deployed memory stores currently operate under retention/deletion mechanisms: a decay factor, an invalidation timestamp, a learned forget gate, or an explicit DELETE call. Despite these mechanisms, forgetting remains a critical failure mode that standard benchmarks often fail to adequately measure [Yang, 2026, Uddin et al., 2026]. A programmatic policy is not a fundamental physical law. For instance, one cannot read the half-life of a stored value off a trained LSTM, identify the exact mechanism that tunes this half-life, or verify the latent mathematical residue once a value is removed.

This gap in fundamental understanding is widest at the point of data removal. Contemporary agent-memory research supplies robust mechanisms for writing, consolidating, and retrieving data. However, for deletion, the standard approach relies on superficial bookkeeping, such as a timestamp or a dropped row [Rasmussen et al., 2025, Chhikara et al., 2025], which leaves measurable residue

within the network architecture even after the entry is nominally deleted [Chakrabortii et al., 2026, Wang and Zhang, 2026]. Further related work is presented in App. B

In this work, we propose and analyze a memory framework where forgetting is a prescribed dynamical property of the store, governed by parameters that can be set, targeted, and analyzed in closed form, rather than an emergent side effect or an external bookkeeping rule. By modeling the store as a physical system, we provide computable answers to three core operational questions:

1. What is the exact retention policy, and where does its optimum lie? (Sec. 2.1)
2. Can retention be specifically scoped to a single localized item? (Sec. 2.2)
3. Is structural deletion truly absolute, and what information mathematically leaks? (Sec. 2.3)

Methodological Foundations. The memory store is constructed as a physical system. The latent state space (q, p) advances via a damped symplectic step of a learnable Hamiltonian, defined as $H = T(p) + V_\theta(q)$. The damping mechanism, defined as $p \mapsto (1 - \gamma)p$, inherently supplies the forgetting function, while Langevin noise introduces the temperature T . Memory items can optionally be written as *atoms*: localized energy wells superposed into a singular energy function that is subsequently read by dynamical relaxation. Our reference architecture is the Causal Learning Unit (CLU), introduced as CHLU by Jawahar and Pierini [2026]. The core of this unit is exactly solvable: when diagonalized within the mass-whitened Hessian, a single damped step acts on each normal mode as a 2×2 matrix entirely fixed by $(\varepsilon\mu, \gamma)$. The spectral radius of this matrix dictates that specific mode’s retention capacity.

Scope and Assumptions. To clarify the boundaries of our empirical claims, we define the following operational assumptions and constraints:

- **Scale:** These early results rely on a novel, designed store. This is not a deployed, large-scale LLM agent memory, nor a generalized system benchmark.
- **Claim Classification:** We report a fundamental mechanism bounded by measured physical laws, not an end-to-end system performance result.
- **Nomenclature borrowing from and building on CHLU Jawahar and Pierini [2026]:**
 - *Inertial mass* (M_i): The learned diagonal of the kinetic term, where larger values result in slower dynamics.
 - *Spectral mass* (μ_k^2): Defined as $\lambda_k(M^{-1/2}\nabla^2 V_\theta M^{-1/2})$, where larger values indicate shorter memory retention. Retention claims always utilize μ .
 - *Stored direction*: A written value occupies a near-flat coset direction.
 - *Integration step* (ε): Fixed at 0.05.
- **Evaluation Metrics:** Half-lives ($n_{1/2}$) are presented with their corresponding read tolerances (Δ) and the thermal persistence length ($\ell_\theta = \varepsilon\sqrt{T/M}/(\gamma r_*)$, the coset register’s physical resolution limit at temperature T).
- **Thermodynamic Consistency:** $T > 0$ strictly requires the fluctuation–dissipation-consistent noise scale $\sigma_i^* = \sqrt{M_i T \gamma (2 - \gamma)}$ utilized alongside a Newtonian kinetic mode.

2 Results

2.1 The Retention Dial Possesses a Computable Optimum

Friction is not a monotone retention knob: retention half-life is non-monotone in relation to friction, so the dial has an optimum rather than a direction. The turning point is fixed by the stored direction’s spectral mass, meaning the optimal setting can be analytically predicted rather than empirically tuned. To understand the macro-dynamics of memory decay, we analyze damping as that retention dial.

The integration step is defined as conformally symplectic: $\det J = (1 - \gamma)^d$, and scales to $(1 - \gamma\phi(q'))^d$ when position-gated (this is derived in App. G.2, and the closed forms below in App. G). Our evaluation utilizes checkpoints comprising a dimension of 4, a hidden size of 64, $\varepsilon = 0.05$, spanning 150 epochs. We assess both **designed units** (featuring an $SO(2)$ -invariant V_θ , across 5 seeds) and **emergent units** (utilizing an MLP V_θ trained on identical symmetric data, across 3 seeds).

84 The latch, the overdamped register, and the underdamped working memory are three specific opera-
 85 tional regimes of a unified curve evaluated at two disparate values of μ . On a fully trained, designed
 86 $SO(2)$ vacuum, the massive radial mode’s retention half-life $n_{1/2}(\gamma)$ forms a V-curve, minimized at
 87 $\gamma_{\text{crit}} = 2\varepsilon\mu_{\text{rad}}$ (Fig. 5, App. C). The overdamped branch tracks the μ^{-2} law until $\varepsilon\mu \approx \gamma/2$, after
 88 which it saturates at a mass-independent floor. This yields log-slopes of -1.006 and between $+1.23$
 89 and $+1.27$ across the 5 validation seeds (See App. C for results). The flat coset (characterized by a
 90 Hessian $\mu^2 \approx 10^{-15}$) maps to the exact same curve at the limit $\mu \rightarrow 0$, where $\gamma_{\text{crit}} \rightarrow 0$.

91 This fundamental law transfers directly to emergent units. Evaluating three emergent MLP check-
 92 points, the coset-tracked $T = 0$ retention half-life — read on the near-flat stored direction that holds
 93 a written value, not on the stiff radial mode of the previous paragraph, though one law governs
 94 both (Fig. 1) — traces the identical V-curve framework: an argmin in units of γ_{crit} , and respective
 95 log-slopes on the two asymptotic branches, across all 3 seeds (Figure 7), with the measured values
 96 in Table 1. Spanning both architecture families, this curve holds consistently across a μ^2 span of
 97 eleven orders (Table 1, Fig. 1). The low endpoint of that span is the ring-profile probe’s resolution on
 98 a checkpoint whose Hessian μ^2 is machine zero rather than a spectral mass, so eleven orders is one
 99 curve on one instrument (probe-floor tick marked in Fig. 6). Put plainly: the curve a designed unit
 100 follows is the curve a learned one follows, read at a different spectral mass.

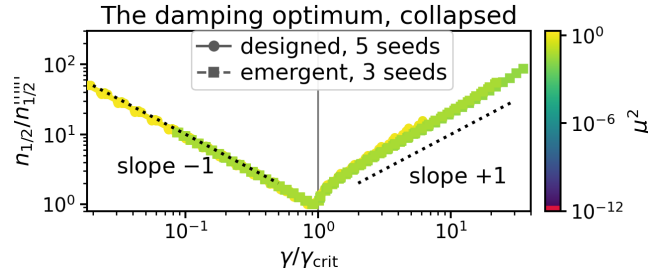


Figure 1: *The retention dial and its optimum, collapsed (headline)*. Eight measured curves collapse onto one: five designed radial modes (circles, verification, $\mu_{\text{rad}}^2 = 0.670\text{--}1.348$) and three emergent learned-MLP coset modes (squares, evidence, $\mu_{\text{soft}}^2 = 2.0\text{--}5.4 \times 10^{-2}$, 3/3 seeds), against the ∓ 1 asymptotes; the designed flat coset is the $\mu \rightarrow 0$ corner of the same curve. (*One-step Jacobian; dim 4, hidden 64, $\varepsilon = 0.05$, single-CPU; langevin_noise="fdt", Newtonian kinetic mode.*)

101 **Cross-Instrument Verification.** Because the primary measurements rely on one-step Jacobians,
 102 we re-measured the entire curve utilizing direct nonlinear rollout, extracting $n_{1/2}$ directly from the
 103 decay envelope’s structural rate. The underlying shape is reproduced well on this second instrument
 104 (see Fig. 10), the two argmins agreeing to 0.35% in units of γ_{crit} (both readings in Table 1). This
 105 establishes that the gap between instruments dictates the level, not the fundamental law (see App. D).

106 2.2 Scoping Retention: Friction Preserves, Temperature Erases

107 At the macro level, isolating a retention policy to a singular entry requires manipulating the physical
 108 dial locally. At absolute zero ($T = 0$), a designed coset latches indefinitely across every friction
 109 setting we evaluated (Fig 5b; drift bound and γ range in Table 1). Conversely, at $T > 0$, the coset
 110 degrades via diffusion, and introducing friction decelerates this decay.

111 We establish that the diffusion coefficient of the stored coordinate is modeled as $D_\theta = \varepsilon T(2 -$
 112 $\gamma)/(2F^2\gamma)$, where the coset scale $F^2 = Mr_*^2$ is the mass-weighted squared displacement of the
 113 latent state per unit stored coordinate (App. G.6), which we verified over a grid of (γ, T) cells
 114 on the designed testbed (Table 1). Consequently, $n_{1/2} \propto \gamma^{+1}$. This sign flip ($\partial n_{1/2}/\partial \gamma > 0$)
 115 holds universally across all evaluated seed and temperature combinations (Figure 4; diffusion-
 116 law diagnostics in Figure 3, Appendix C). Only temperature exhibits an inverse erasure effect
 117 ($\partial n_{1/2}/\partial T < 0$).

118 **The Memory Vault Mechanism.** We propose the memory vault: a position-gated friction field that
 119 is absorb-only. A localized spatial hole within this field acts simultaneously as a brake, increasing
 120 dissipation, and a refrigerator, reducing the local effective temperature. This mechanism generates a
 121 retention vault factor of $(\gamma_{\text{eff}}/\gamma)^2$ on designed architecture (Fig. 2), read from the $\hat{D}_\theta$ estimator and

122 quoted against a uniform-scalar-friction control at matched γ_{eff} ; the direct first-passage reading is
 123 boundary-layer biased on the outside arm ($\ell_\theta/\Delta = 0.079$) and is therefore not the quoted vault. All
 124 four readings — the $\hat{D}_\theta$ vault, the scalar control, the first-passage value, and the local temperature —
 125 are in Table 1. Crucially, this localized field mathematically confines the register: the state fraction
 126 outside the boundary ($|\theta| > 1$ rad) drops definitively to 0.0000 within the hole (see Fig. 11 and
 127 App. E). Put plainly: the vault factor is how much longer a written value is retained inside the hole
 128 than outside it.

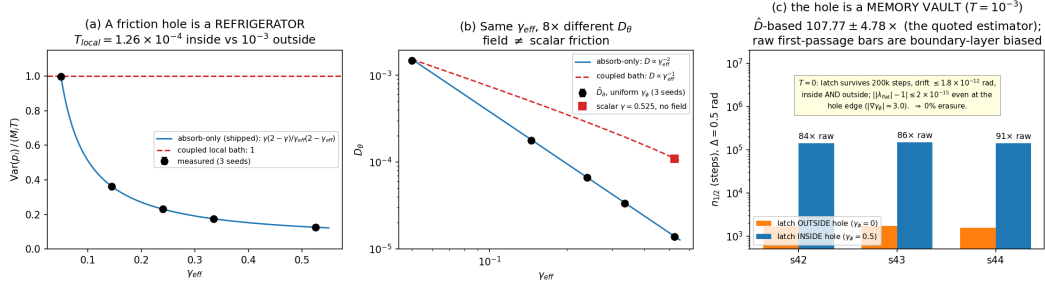


Figure 2: The friction hole as a memory vault, on the designed arm. (a) The hole is a refrigerator: the momentum variance follows the absorb-only curve and rejects the coupled-bath prediction, giving $T_{\text{local}} = 1.26 \times 10^{-4}$ inside against 10^{-3} outside. (b) At the same γ_{eff} the diffusion constant differs by $8\times$ between a friction field and a scalar friction, which is the mechanism statement in one panel. (c) The vault factor: $107.77 \pm 4.78\times$ from the $\hat{D}_\theta$ estimator; the per-seed bars beneath it are raw first-passage ratios, boundary-layer biased on the outside arm, which is why the pooled first-passage reading of $86.97 \pm 2.94\times$ is not quoted as the vault. The inset states the $T = 0$ control: the hole erases nothing. *Designed $SO(2)$ with an analytic field, 3 seeds, $T = 10^{-3}$, $\Delta = 0.5$ rad, $\text{langevin_noise} = \text{"fdt"}$, Newtonian kinetic mode, laptop-CPU: verification.*

129 2.3 Structural Deletion: Absolute Guarantees, Lifetime Trade-offs, and Residual Leakage

130 In standard architectures, deleting an item from memory relies on stochastic, best-effort heuristics,
 131 such as dropping active rows or writing system tombstones. We evaluate whether the store’s intrinsic
 132 physical state can be reduced to a pure function of its live set alone — that is, whether the layout
 133 depends only on which items are currently stored, not on the order or history of the writes that
 134 stored them. When data points are continuously superposed into a singular energy function, standard
 135 placement becomes inherently path-dependent.

136 To resolve this, we map memory placement canonically, ensuring it remains an exclusive function of
 137 the live records and the predefined geometry. Utilizing Blleloch and Golovin’s [2007] stable-matching
 138 table rules, we establish exact store-level deletion. Our measurements (see App. F) confirm that with
 139 the waitlist engaged, store-level deletion remains exact across all evaluated loads ($0.29\times$ to $1.71\times$
 140 of standard lattice capacity). Under explicit conditions (a budget sufficient for the cell count, zero
 141 baseline leakage, and priority/attribute-based eviction; recency-based eviction remains intrinsically
 142 history-dependent and is excluded), the physical layout of the store remains byte-identical to a
 143 temporal state that theoretically never encountered the deleted item. This is a store-level guarantee
 144 only; the frozen encoder and any residue of past writes in a learned landscape are separate channels,
 145 and we claim no certified (ε, δ) unlearning.

146 **The Lifetime Trilemma and Residual Leakage:** Memory lifetimes are governed by a strict
 147 trilemma: exact value fidelity, amplitude-independent address hold, and amplitude-independent read
 148 latency. A system may only optimize two. We also note that physical decay provides architectural
 149 retrieval geometry rather than cryptographic privacy (App. F).

150 3 Limitations and Future Work

151 To define the scope and bounds of our empirical evidence, we identify the following limitations and
 152 subsequent future pathways:

- 153 • **Scale and Architecture Scope:** This research is strictly constrained to a dimension of 4,
154 a hidden size of 64, and small-scale execution on a single CPU across 5 designed and 3
155 emergent seeds. No larger-scale performance should be inferred and this scaling is left to
156 future work.
- 157 • **The Designed-Symmetry Precondition:** The exact $T = 0$ retention latch currently requires
158 precise continuous coset geometries that must be designed into the architecture.
- 159 • **Mandatory Thermodynamic Scope:** The raw T/γ -exponents cannot distinguish between
160 designed and emergent units. Evaluating any state where $T > 0$ requires the strict combina-
161 tion of fdt noise and Newtonian kinetic modes.
- 162 • **Deletion Scope:** The exact deletion guarantee applies strictly at the isolated store-level
163 beneath set-function eviction models. Furthermore, recency-based eviction intrinsically
164 breaks latent history independence and is thus excluded.
- 165 • **No Task-Level Results:** This work presents a theoretical concept with empirical validation.
166 Testing on specific task-level benchmarks is left to future work.
- 167 • **Future Work Directions:** We intend to evaluate wider emergent collapses in μ^2 , remeasure
168 the memory vault contrast, formally assess the amortized per-delete execution cost against
169 flat datastore architectures, and construct localized temperature fields.

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267 A Values Quoted in the Main Text

268 Sec. 2 states its results in words; the measured values behind them are collected here, each beside the
 269 arm it was measured on and the rider it must be quoted with. *Arm* names the class of the measurement:
 270 **verification** = a *designed testbed* (an architecturally designed $SO(2)$ potential), **evidence** = a *learned*
 271 *system* (an MLP V_θ trained on symmetric data). Every value is repeated from the section or appendix
 272 named beside it; none is new here.

273 B Extended Related Work

274 **Retention Policies in Deployed Memory Systems:** Agent memories operate as external data
 275 stores that a model actively writes to, retrieves from, and dynamically edits (e.g., MemGPT, Packer
 276 et al., 2024; Mem0, Chhikara et al., 2025). Alternatively, bounded compressive states exist internal
 277 to the network (e.g., Infini-attention, Munkhdalai et al., 2024). The retention policies governing
 278 these systems rely on scores and gates. Examples include a 0.995 recency decay factor in ranking
 279 heuristics [Park et al., 2023], Ebbinghaus decay functions applied to memory strength [Zhong et al.,
 280 2024], learned per-memory expiration spans [Sukhbaatar et al., 2021], and dynamic forget gates α_t
 281 optimized during test time [Behrouz et al., 2025]. While the learned gate in Titans [Behrouz et al.,
 282 2025] is structurally similar to the retention dial discussed in Sec. 2.1, none of these prior works
 283 establish physical retention laws, state explicit half-lives, define read tolerances, or mathematically
 284 price the information loss.

285 **Deletion Mechanisms: Deployed and Formal:** Current deployment architectures handle deletion
 286 via invalidation. For example, Zep invalidates contradicted edges with timestamps while preserving
 287 the latent record [Rasmussen et al., 2025], and Mem0 simply drops rows from its active context.
 288 Conversely, exact byte-identity is crucial for security, as soft-deleted vectors in graph ANN databases
 289 remain reconstructible from raw index files [Chakrabortii et al., 2026]. Recent benchmarking
 290 highlights that real-world deployment failures are predominantly forgetting failures [Yang, 2026],
 291 with outdated retention driving significant recommendation errors [Uddin et al., 2026]. Formally,
 292 *certified* removal is Guo et al.’s [2020] Sec. 2, Eq. (1), an ε condition with an unnumbered (ε, δ)
 293 relaxation immediately after; exact methods delete by isolation, with cost and capacity formalisms in
 294 place [Bourtole et al., 2019, Ginart et al., 2019, Sekhari et al., 2021]. We sit functionally between
 295 these approaches: we offer a store-level, bit-exact structural guarantee based on canonical geometric
 296 placement [Blelloch and Golovin, 2007], rather than a stochastic system-level unlearning heuristic.

Table 1: The measured values behind Sec. 2, with the arm each was measured on and the rider each must travel with. *Arm: verification* = designed testbed; *evidence* = learned system. All rows come from the checkpoints and grids of Sec. 2.1 and Apps. C–E (dim 4, hidden 64, $\varepsilon = 0.05$, single-CPU); every $T > 0$ cell uses `langevin_noise="fdt"` with a Newtonian kinetic mode.

Result	Quantity	Value	Arm	Scope / rider
V-curve, emergent arm (Sec. 2.1, App. D)	argmin, one-step Jacobian	$0.902 \pm 0.003 \times \gamma_{\text{crit}}$	evidence	3 emergent MLP seeds, $T = 0$; a learned potential, not a designed one
	log-slope, over-damped branch	-1.0020 ± 0.0003	evidence	same 3 emergent seeds
	log-slope, under-damped branch	$+1.116 \pm 0.011$	evidence	same 3 emergent seeds
Collapse (Sec. 2.1)	span μ^2 range spanned by the one curve	$[1.7 \times 10^{-12}, 7 \times 10^{-2}]$	both arms	Probe resolution: the low endpoint is the ring-profile probe’s resolution on a checkpoint whose Hessian μ^2 is machine zero — <i>not</i> a measured spectral mass
Cross-instrument check (App. D)	argmin, rollout envelope rate	$0.9001 \pm 0.0052 \times \gamma_{\text{crit}}$	evidence	3 emergent seeds, $\delta = 0.05$ rad
	argmin, one-step Jacobian	$0.9032 \pm 0.0027 \times \gamma_{\text{crit}}$	evidence	same cells; the instrument gap is a level, not a rate
$T = 0$ latch (Sec. 2.2, App. C)	coset drift	$\leq 4.9 \times 10^{-12}$ rad / 200k steps	verification	designed $SO(2)$ coset, every $\gamma \in [0.002, 0.5]$; $T = 0$ only
Coset diffusion law (Sec. 2.2, App. C)	$\hat{D}_\theta / D_\theta^{\text{pred}}$	1.0068 ± 0.0219	verification	25 (γ, T) cells on one trained designed checkpoint (seed 44), $\Delta = 0.5$ rad
Friction-hole vault (Sec. 2.2, App. E)	T_{local} inside vs. outside	1.26×10^{-4} vs. 10^{-3}	verification	designed arm, $\gamma = 0.05$, $\gamma_\phi = 0.5$; absorb-only field
	vault factor, $\hat{D}_\theta$ estimator	$107.77 \pm 4.78 \times$	verification	the quoted vault; it travels with this estimator’s name. Designed $SO(2)$, 3 seeds, $T = 10^{-3}$, $\Delta = 0.5$ rad
	scalar-friction control	$13.28 \pm 0.12 \times$	verification	uniform scalar friction at matched γ_{eff} , same cells
	raw first-passage ratio	$86.97 \pm 2.94 \times$	verification	boundary-layer biased on the outside arm ($\ell_\theta / \Delta = 0.079$); <i>not</i> the quoted vault

297 **Physical and Learned Forgetting Laws:** While learned recurrent models like LSTMs [Hochreiter
298 and Schmidhuber, 1997] lack stateable physical forgetting laws, biological and artificial representa-
299 tional drift has been observed to reorganize internal unit representations over time [Aitken et al., 2022].
300 Our work differs by tracking a specific coset coordinate under strictly defined physical dynamics.
301 In theoretical physics, dissipation transitions type-A Nambu–Goldstone modes from propagating to
302 diffusive states [Minami and Hidaka, 2018]. Furthermore, strictly equivariant flows possess Lyapunov
303 neutral modes [Mo, 2026] that provide kinematic protection, a gap complementary to our established
304 physical budget (μ^2).

305 C The effects of (μ, γ, T) : The Diffusion Law and the Sign Flip

306 To ensure complete rigor, all metrics detailed here derive from the designed $SO(2)$ testbed, providing
307 strict verification. Furthermore, all $T > 0$ cells mandate the explicit use of `langevin_noise="fdt"`
308 operating within a Newtonian kinetic mode. The legacy reference default structure fundamentally
309 violates these thermodynamic laws. Every reported half-life $n_{1/2}$ is paired unconditionally with its
310 specific read tolerance $\Delta = 0.5$ rad and its ratio ℓ_θ / Δ .

311 **Diffusion Law Diagnostics (25 Cells):** We evaluate a discrete $5\gamma \times 5T$ grid utilizing seed 44. The
312 ratio $\hat{D}_\theta / D_\theta^{\text{pred}}$ yields 1.0068 ± 0.0219 , bounded between a minimum of 0.9644 and a maximum of
313 1.0484. The gradient $d \log D / d \log T$ records values of +0.998, +0.997, +0.992, +1.010, +1.006.

314 Simultaneously, the gradient $d \log D / d \log \gamma$ evaluates to $-1.014, -0.980, -0.999, -1.001, -0.982$
 315 following the mathematical isolation of the discrete $(2 - \gamma)$ factor.

316 **Sign Flip Verification:** We observe $d \log n_{1/2} / d \log \gamma = +0.9552 \pm 0.0422$ at a baseline tempera-
 317 ture of $T = 10^{-3}$ across all 5 testing seeds, presenting positively in all 10/10 examined conditions.
 318 The measured effect size for the transition $\gamma : 0.05 \rightarrow 0.2$ results in a precise $3.77 \pm 0.23 \times$ in-
 319 crease ($\ell_\theta / \Delta < 0.05$; seed-specific values: 3.55, 3.64, 3.61, 3.89, 4.15). The inverse thermal driver
 320 measures at $d \log n_{1/2} / d \log T = -0.956$ to -0.979 within the deep-diffusive operational bounds.

321 **Designed $T = 0$ Latch Properties:** Evaluating absolute zero, the metric $||\lambda_{\text{flat}}| - 1| \leq 1.7 \times 10^{-14}$
 322 across 22 specific γ values on 5 independent seeds. Consequently, the observed drift is restricted to
 323 $\leq 4.9 \times 10^{-12}$ rad per 200k integration steps across 30/30 evaluated cells. This strongly contrasts
 324 with the $\gamma = 0$ baseline control, which incurs a massive 142.7 rad drift over a significantly shorter
 325 20k steps.

326 **Designed V-Curve Minimization:** The optimal minimum argmin lies between 0.076 and 0.096,
 327 against the computationally predicted bounds of 0.082–0.116 (utilizing per-seed specific values
 328 $\mu_{\text{rad}}^2 = 0.670, 0.771, 1.190, 1.093, 1.348$). The resulting asymptotic log-slopes measure -1.006 and
 329 between $+1.23$ and $+1.27$, enforcing a mass-independent floor of exactly 27.03 steps measured at
 330 $\gamma = 0.05$.

Table 2: The $T > 0$ face of the budget, all four entries continuously measured on the designated designed testbed.

Stored Direction	$\partial n_{1/2} / \partial \gamma$ (Increased Friction)	$\partial n_{1/2} / \partial T$ (Increased Heat)
Coset / flat (de- signed)	> 0 , friction <i>preserves</i> ($+0.955 \pm 0.042$; $3.77 \times$ at $\gamma : 0.05 \rightarrow 0.2$)	< 0 , temperature <i>erases</i> (-0.956 to -0.979)
Massive (under- damped)	< 0 , friction shortens (slope -1.006)	≈ 0 ; T sets only the fluctuation floor (max/min ≤ 1.013)

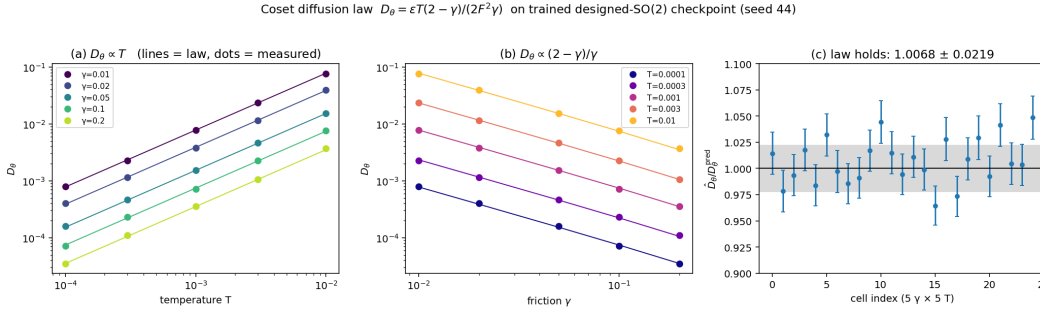


Figure 3: The coset diffusion law, detailed systematically panel by panel. (a) $D_\theta \propto T$ measured precisely at five independent frictions; (b) $D_\theta \propto (2 - \gamma)/\gamma$ evaluated at five specific temperatures; (c) the absolute ratio $\hat{D}_\theta / D_\theta^{\text{pred}}$ compiled over all 25 structural cells, reading exactly 1.0068 ± 0.0219 alongside the respective drawn band. *Multi-seed status: evaluated utilizing one fully trained designed-SO(2) checkpoint (seed 44). langevin_noise="fdt", Newtonian kinetic mode, $\Delta = 0.5$ rad. Designed testbed: validation of verification.*

331 D The Emergent Arm, and the V-Curve on a Second Instrument

332 We evaluate the emergent MLP checkpoints utilizing 3 unique seeds alongside a precisely matched
 333 designed control. All predictions detailed below were structurally filed and pre-registered prior to the
 334 execution of any test harness, establishing rigorous evidence parameters.

335 We deploy four specific measurement instruments over identical fit windows (defined as $\gamma < \gamma_{\text{crit}}/2.5$
 336 and $\gamma > 2.5\gamma_{\text{crit}}$): 1. *I-J*: The exact one-step Jacobian, where $n_{1/2} = \ln 2 / (-\ln |\lambda_{\text{ret}}|)$. 2. *I-RI*:

Friction preserves, temperature erases (trained designed-SO(2) checkpoint, seed 44)

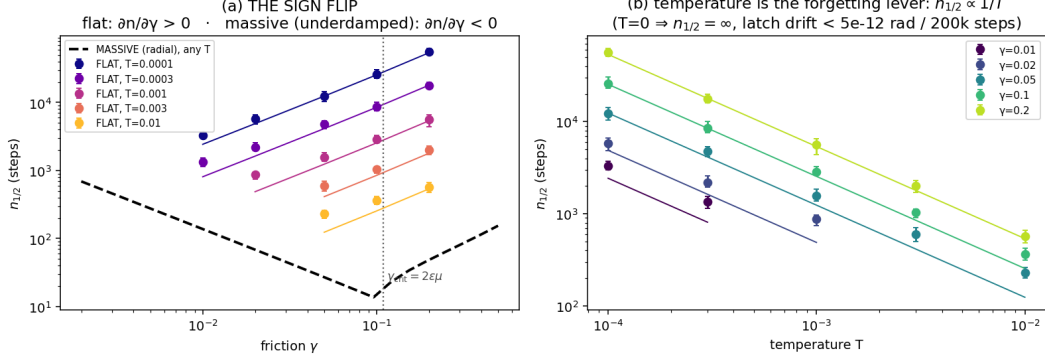


Figure 4: The sign flip, and the lever that does erase. (a) The flat coset’s half-life *rises* with friction at every temperature, while the underdamped massive mode’s falls — the two stored directions respond to the dial with opposite sign, with $\gamma_{\text{crit}} = 2\epsilon\mu$ marked. (b) Temperature is the forgetting lever, $n_{1/2} \propto 1/T$ at every friction. *Multi-seed status: one trained designed-SO(2) checkpoint (seed 44); the 10/10 seed $\times$ temperature statement and the $+0.9552 \pm 0.0422$ slope are this appendix’s five-seed numbers, not this figure’s. langevin_noise="fdt", Newtonian kinetic mode, $\Delta = 0.5$ rad. Designed testbed: verification.*

The two regimes have different damping physics (exact one-step Jacobian at the vacuum)

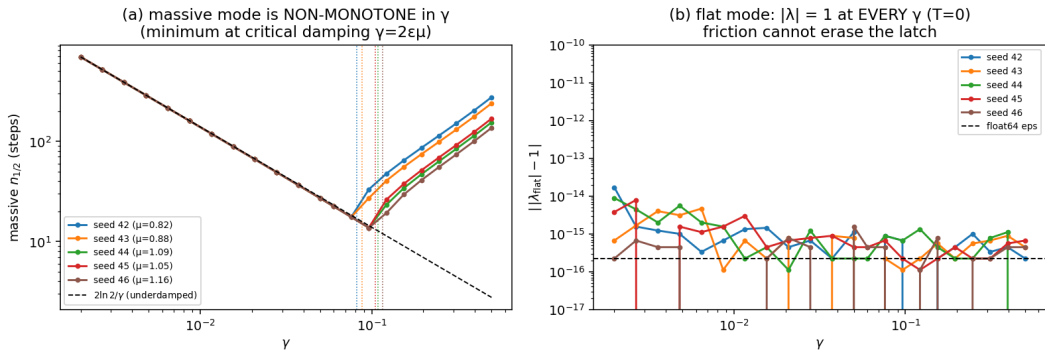


Figure 5: The two regimes, read off the exact one-step Jacobian at the vacuum. (a) The massive radial mode is non-monotone in γ , each seed’s minimum sitting at its own $\gamma_{\text{crit}} = 2\epsilon\mu$ (dotted), against the dashed underdamped envelope $2 \ln 2/\gamma$ drawn on the same axes. (b) The flat mode holds $|\lambda| = 1$ at every γ to the float64 floor: friction cannot erase the latch. *Five trained designed seeds, $T = 0$, dim 4, hidden 64, $\epsilon = 0.05$, laptop-CPU: verification.*

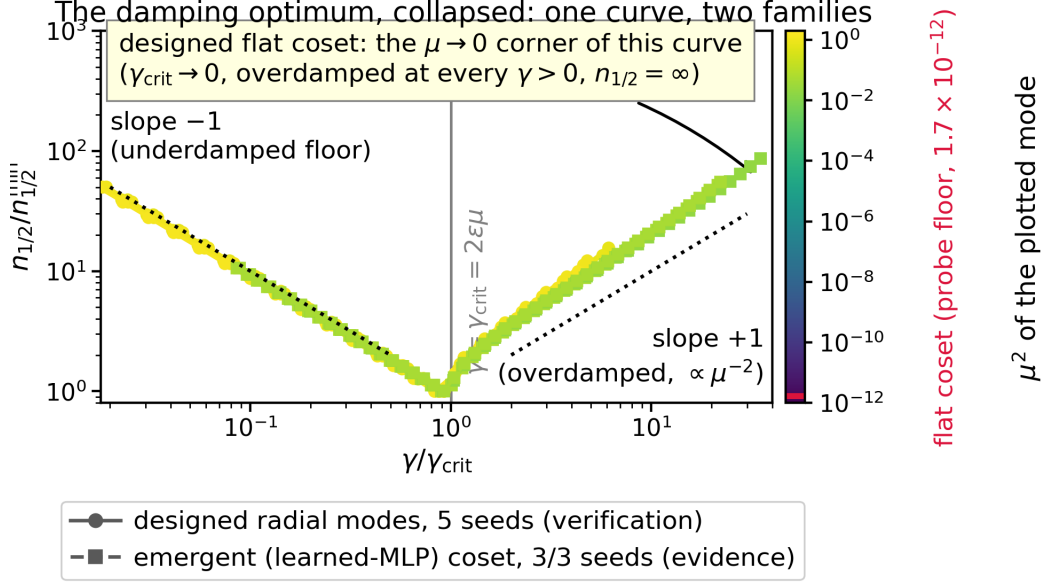


Figure 6: *The damping optimum, collapsed: the full-size version of Figure 1.* The same eight measured curves and the same ± 1 asymptotes, carrying the annotations the main-text version omits: the two-family legend, the $\gamma = \gamma_{\text{crit}} = 2\varepsilon\mu$ marker, the $\mu \rightarrow 0$ flat-coset callout, and the crimson probe-floor tick on the μ^2 colorbar (1.7×10^{-12}). No plotted value differs from Figure 1. (*One-step Jacobian; dim 4, hidden 64, $\varepsilon = 0.05$, laptop-CPU; $\text{langevin_noise} = \text{"fdt"}$, Newtonian kinetic mode.*)

337 The rollout first crossing threshold of $|\theta| \leq \delta/2$. 3. *I-R2*: The rollout last crossing threshold. 4. *I-R3*
338 (Primary): The rollout envelope-rate fit evaluated specifically over the late window, guaranteeing
339 $R^2 \geq 0.9955$ (median 0.9996) across every evaluated cell.

Table 3: The foundational shape claims evaluated on all four designated instruments ($\delta = 0.05$ rad, exactly 48 γ values, utilizing 3 emergent seeds). The rollout rate instrument reproduces the Jacobian; both threshold instruments fail below γ_{crit} , so no rollout $n_{1/2}$ is ever quoted from a threshold there. Rows are stacked by quantity so that no column is truncated; no value differs from the per-instrument reading.

Quantity	Instrument	Seed 42	Seed 43	Seed 44	Mean $\pm$ sd
γ_{crit}	—	0.02334	0.01424	0.02265	—
Argmin/ γ_{crit}	I-J	0.8994	0.9046	0.9055	0.9032 ± 0.0027
	I-R1	0.0857	0.1404	0.0883	0.1048 ± 0.0252
	I-R2	0.2068	0.1404	0.3031	0.2168 ± 0.0668
	I-R3	0.8928	0.9047	0.9028	0.9001 ± 0.0052
Slope Below	I-J	−1.0023	−1.0016	−1.0022	$−1.0020 \pm 0.0003$
	I-R1	+0.0725	+0.0688	+0.0455	$+0.0623 \pm 0.0120$
	I-R2	−1.0556	+0.0688	−0.8854	$−0.6240 \pm 0.4948$
	I-R3	−0.9651	−0.9977	−0.9854	$−0.9827 \pm 0.0134$
Slope Above	I-J	+1.1262	+1.1031	+1.1254	$+1.1182 \pm 0.0107$
	I-R1	+1.1027	+1.0889	+1.0977	$+1.0964 \pm 0.0057$
	I-R2	+1.1027	+1.0889	+1.0977	$+1.0964 \pm 0.0057$
	I-R3	+1.1255	+1.1030	+1.1231	$+1.1172 \pm 0.0101$

340 **Instrument Level Disparities:** On the strictly overdamped branch ($\gamma \geq 2\gamma_{\text{crit}}$), the physical
341 decay rates match perfectly, whereas the *threshold* instrument is consistently offset by an exact per-
342 checkpoint constant flat in relation to γ . The full-window value for Seed 44 measures 0.9216 ± 0.0027 ,

343 sitting precisely 0.0004 beneath its pre-registered bounds due to the residual presence of a secondary
 344 faster mode within the window. The primary cause of this threshold offset is confirmed: the ring write
 345 mechanism consistently deposits only a specific limited fraction of its absolute amplitude directly
 346 into the designated slow branch (envelope-fit intercept over δ measuring 0.638/0.644/0.327).

Table 4: The instrument gap is a level, not a rate: on the overdamped branch ($\gamma \geq 2\gamma_{\text{crit}}$) the decay *rates* agree while the *threshold* instrument is offset by a per-checkpoint constant flat in γ . Seed 44’s full-window value (0.9216 ± 0.0027) sits 0.0004 below its pre-registered band because that window still contains a second, faster mode (early-window ratio 0.539); the window-split diagnostic converges monotonically to 1 into the asymptote. Cause of the threshold offset, measured: the ring write deposits only a fraction of its amplitude in the slow branch (envelope-fit intercept over $\delta = 0.638/0.644/0.327$).

seed	$\Gamma_{\text{jac}}/\Gamma_{\text{R3}}$ (asymptotic win- dow)	$n_{\text{jac}}/n_{\text{R1}}$ (threshold)	CV of $n_{\text{jac}}/n_{\text{R1}}$ over the grid
42	0.9935	1.3307	2.2–3.7% over 21–25 γ -points
43	1.0000	1.7981	
44	0.9948	1.8204	
mean	0.995 ± 0.003	per seed, at $\delta = 0.05$ rad	$ d \log(\text{ratio})/d \log \gamma \leq 0.0034$ (threshold 0.10)

347 **Finite Write Amplitudes:** At the specific values $\delta = 0.05/0.2/0.5$ rad, the designated
 348 I-R3 argmin logs at 0.8928/0.9007/0.8968 (seed 42), 0.9047/0.9045/0.9174 (seed 43), and
 349 0.9028/0.9037/0.9070 (seed 44). This variance accounts for a maximum shift of +1.4% across a
 350 complete order of magnitude ($10\times$ range), with identical slope variances constrained strictly beneath
 351 1.9%. Every iteration at $\delta = 0.5$ relaxed identically to absolute baseline ($|\theta_{\text{final}}| \leq 6.5 \times 10^{-10}$),
 352 preventing any seed’s write function from clearing its designated washboard barrier.

353 **Left Branch Exclusions:** For conditions where $\gamma < \gamma_{\text{crit}}$, the respective coset eigenpair funda-
 354 mentally becomes complex with $|\lambda|^2 = 1 - \gamma$. This forces $n_{1/2} = \ln 2 / (-\frac{1}{2} \ln(1 - \gamma))$ to operate
 355 entirely independent of both μ^2 and the specific architectural checkpoint. For instance, at exactly
 356 $\gamma = 0.002$, emergent seeds 42 ($\mu^2 = 5.449 \times 10^{-2}$) and 43 ($\mu^2 = 2.029 \times 10^{-2}$) return identical
 357 metrics: $\lambda_{\text{ret}} = 0.998999499$ alongside an $n_{1/2}$ of 692.5, aligning tightly against the calculated
 358 $2 \ln 2 / 0.002 = 693.1$.

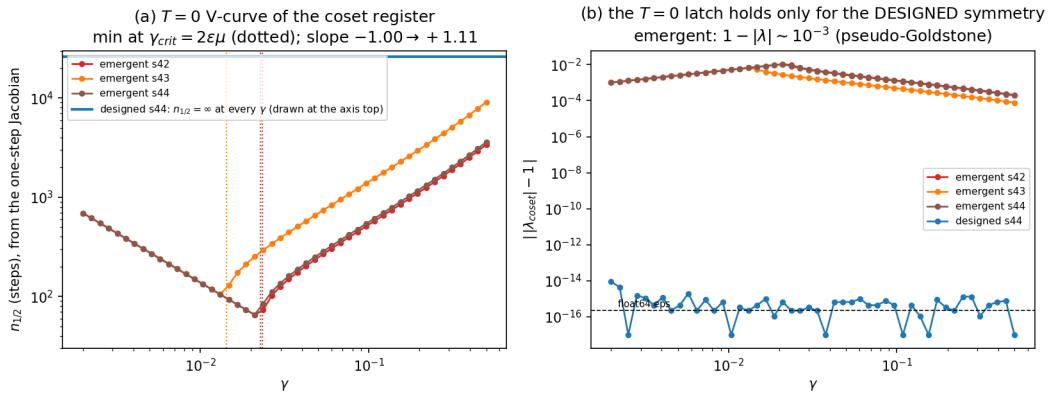


Figure 7: The exact emergent V-curve depicted un-collapsed, outlining the distinct negative baseline parameters beside it. (a) Displays the precise $T = 0$ half-life charted against friction for each separate emergent seed compiled from the one-step Jacobian. (b) Tracks $|\lambda_{\text{coset}} - 1|$, revealing exact floor alignment for the strictly designed symmetry but falling short by precisely $\sim 10^{-3}$ on all emergent seeds. *Evaluated using three emergent MLP seeds against one rigidly matched designed control.*

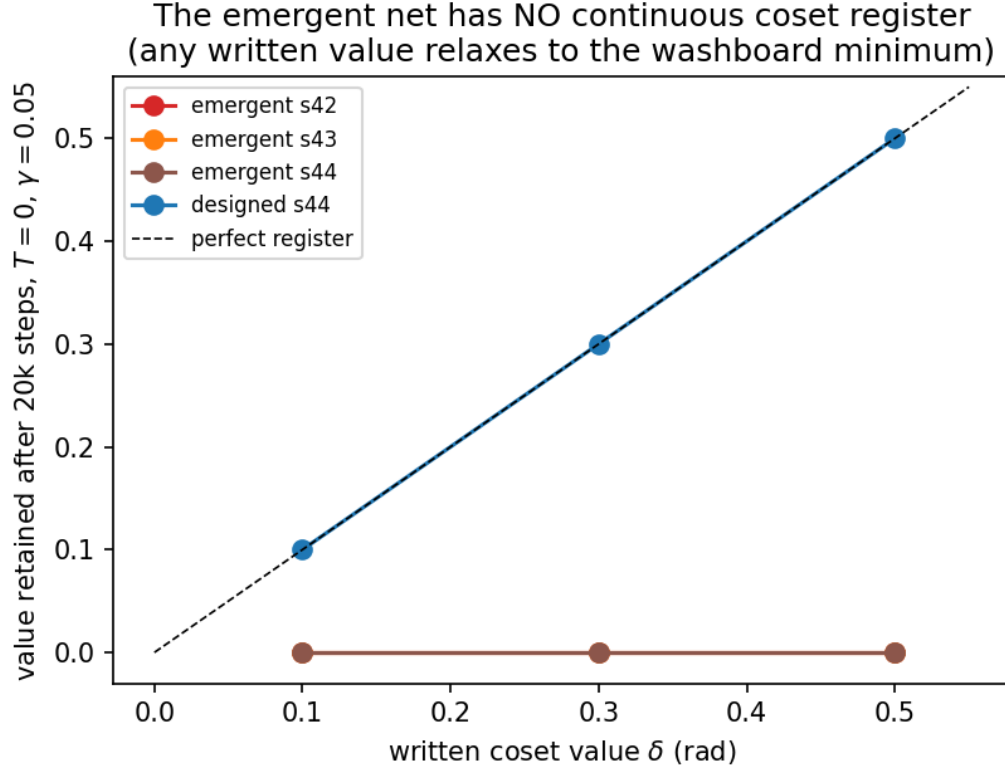


Figure 8: The emergent unit has no continuous coset register. A written coset value δ is held for 20 000 steps at $T = 0$, $\gamma = 0.05$: on every emergent seed the value relaxes to the washboard minimum and nothing is retained, where the designed checkpoint returns δ exactly. *Three emergent MLP seeds plus a matched designed control: evidence.* This is the figure behind the capacity statement of $\approx 1\text{--}1.6$ bits.

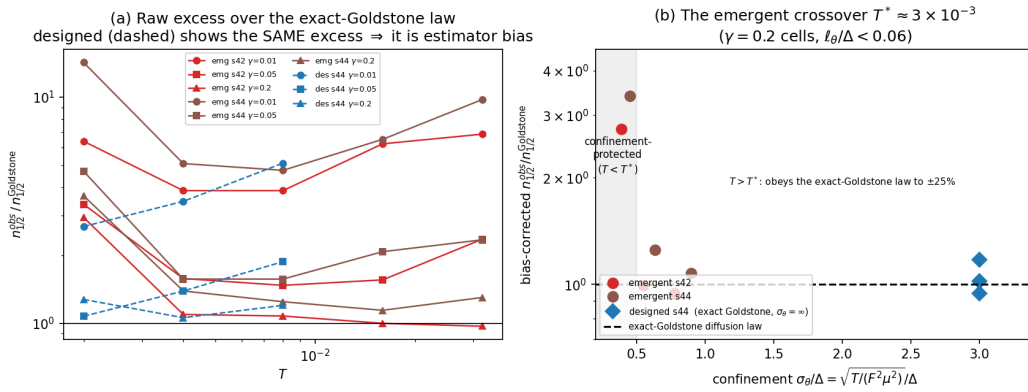


Figure 9: The crossover T^* , and why the raw curve cannot be read directly. (a) The raw excess over the exact-Goldstone law is the same on the matched designed control as on the emergent seeds, so the excess is estimator bias rather than physics. (b) Bias-corrected, the emergent coset joins the exact-Goldstone law above $T^* \approx 3 \times 10^{-3}$, the shaded band being the confinement-protected cells below it. *Multi-seed status, stated plainly: two emergent seeds and one matched designed control, an $n < 3$ cell.* $\Delta = 0.5$ rad, $\ell_\theta/\Delta < 0.06$, langevin_noise="fdt", Newtonian kinetic mode: evidence.

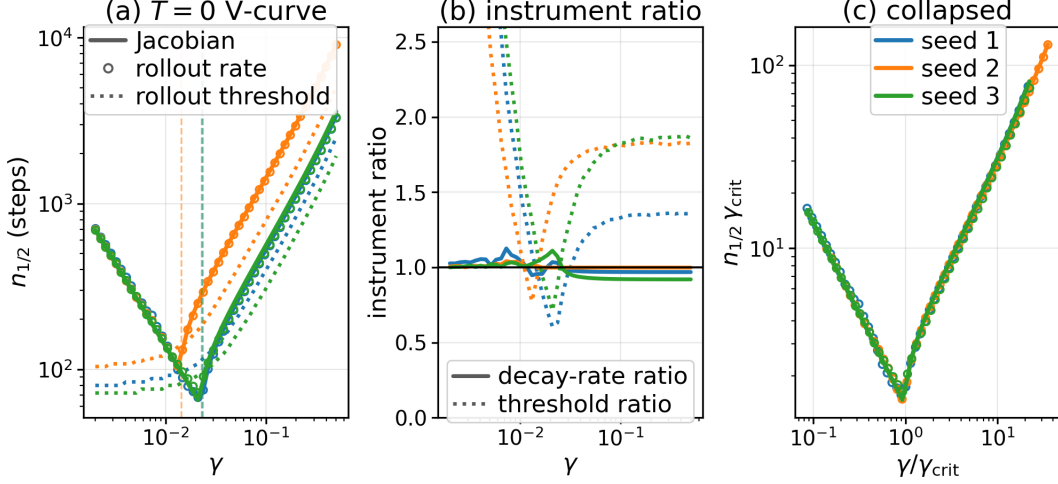


Figure 10: The V-curve measured a second way. (a) $T = 0$ $n_{1/2}(\gamma)$ per emergent seed on the one-step Jacobian (I-J, lines), the rollout envelope-rate instrument (I-R3, circles) and the rollout first-crossing threshold (I-R1, dotted), at $\delta = 0.05$ rad. (b) The per- γ instrument ratio: the rate ratio sits on 1, the threshold ratio is offset and flat in γ above γ_{crit} . (c) The same data in collapsed variables. (*Emergent MLP, 3 seeds, dim 4, hidden 64, $\varepsilon = 0.05$, float64, $T = 0$, pre-registered $\Rightarrow$ evidence.*)

E The Friction-Hole Vault, on Both Arms

This mechanism dictates that the shipped field remains strictly absorb-only, following $p \leftarrow (1 - \gamma_\phi)(1 - \gamma)p + \sigma(\gamma)\xi$ with $\sigma^2 = MT\gamma(2 - \gamma)$. This is dictated strictly by the *scalar* γ , yielding parameters $\gamma_{\text{eff}} = 1 - (1 - \gamma)(1 - \gamma_\phi)$, $T_{\text{local}} = T\gamma(2 - \gamma)/[\gamma_{\text{eff}}(2 - \gamma_{\text{eff}})]$ and importantly $D_\theta = \varepsilon T\gamma(2 - \gamma)/(2F^2\gamma_{\text{eff}}^2)$. The resulting vault effectively computes to $(\gamma_{\text{eff}}/\gamma)^2$. Conversely, a coupled bath alternative would dictate $D_\theta \propto \gamma_{\text{eff}}^{-1}$ returning a vault of $13.88\times$ exactly at $\gamma = 0.05$, $\gamma_\phi = 0.5$. These competing hypotheses scale apart by a factor of $7.942\times$; experimental results comprehensively refute the coupled-bath prediction.

Designed Arm Assessment: We observe $\text{Var}(p_i)/(M_i T) = 0.9986$ at baseline ($\gamma_\phi = 0$), scaling to 0.36171 ± 0.00129 at (0.1), 0.23044 at (0.2), 0.17513 at (0.3), and definitively hitting 0.12600 ± 0.00031 at (0.5). This closely mirrors the pure absorb prediction of 0.12591 . Consequently, T_{local} drops precisely to 1.26×10^{-4} , equating to a $7.94\times$ functioning refrigerator effect. The comprehensive D_θ -based structural vault records exponential factors of $8.42\times$, $22.39\times$, $44.31\times$, and finally $107.77 \pm 4.78\times$ specifically at $\gamma_\phi = 0.1/0.2/0.3/0.5$ (matching the absorb baseline prediction of 110.25). Conversely, the scalar control isolated at identical γ_{eff} reads merely $13.28 \pm 0.12\times$. At true absolute zero ($T = 0$), the hole functionally erases absolutely nothing, logging drift boundaries of strictly $\leq 1.75 \times 10^{-12}$ rad per 200k continuous execution steps inside and out.

Emergent Arm Translation: Since the structural coset here is definitionally bounded ($\mu_{\text{adiab}}^2 = 2.77 \times 10^{-2}$ to 8.93×10^{-2}), the baseline designed block estimator is biased artificially low by nearly 55%. It is therefore systematically replaced by a rigorously cross-validated, window-free Ornstein-Uhlenbeck fit parameter measuring $\text{MSD}(n) = A(1 - e^{-n/\tau})$ and $\hat{D} = A/(2\varepsilon\tau)$. On the strictly bounded evaluation cells, the law-referenced absolute vault computes successfully to $106.1 \pm 5.0\times$ against the fundamental prediction of $110.25\times$. The contrast metric, however, remains uniquely native to the designed architecture: the measured field-to-scalar control variance computes to 23.39 ± 10.06 against a pre-registered bounds parameter of $[6.5, 9.5]$, triggering the designated falsifier constraint strictly because the control arm intrinsically delocalizes while the main field functionally adheres to the absorb limit.

Table 5: The refrigerator law transfers exactly to an emergent register: $\text{Var}(p_i)/(M_i T)$ against the absorb-only prediction, 2048 walkers $\times$ 40 samples, 3 emergent seeds; over all 24 field cells observed/absorb = 0.9998 ± 0.0019 , and at $\gamma_\phi = 0.5$ $T_{\text{local}}/T = 0.12570$, a $7.955\times$ refrigerator against 7.942 predicted. The coupled bath predicts 1.0 everywhere and is measured at 0.2235.

T	γ_ϕ	γ_{eff}	absorb pred.	coupled pred.	measured	obs/absorb
4×10^{-3}	0.0	0.050	1.00000	1.0	0.99815 ± 0.00257	0.9981
4×10^{-3}	0.1	0.145	0.36249	1.0	0.36326 ± 0.00067	1.0021
4×10^{-3}	0.2	0.240	0.23082	1.0	0.23062 ± 0.00080	0.9991
4×10^{-3}	0.3	0.335	0.17480	1.0	0.17473 ± 0.00046	0.9996
4×10^{-3}	0.5	0.525	0.12591	1.0	0.12592 ± 0.00055	1.0001
8×10^{-3}	0.5	0.525	0.12591	1.0	0.12548 ± 0.00032	0.9966

Table 6: The hole confines an emergent register, and the laundering control shows friction alone does not. $T = 4 \times 10^{-3}$, 1024 walkers, burn-in 20 000–80 000 steps, spread by interquartile range; “hop” means $|\theta| > 1$ rad. The pre-registered stationary-spread comparison is *void* here and no in/out spread ratio may be quoted from these checkpoints: the control that must return exactly 1.000 returns 0.4586 ± 0.1181 .

seed	arm	γ_{eff}	σ_θ (IQR)	hop fraction	Boltzmann at T_{local}	obs/pred
42	no hole, $\gamma = 0.05$	0.050	0.40795	5.50%	0.24073	1.6946
42	scalar 0.525 (control)	0.525	0.20034	0.73%	0.24073	0.8322
42	γ_ϕ hole	0.525	0.06442	0.0000	0.08542	0.7542
43	no hole	0.050	1.17601	42.97%	0.46059	2.5533
43	scalar 0.525	0.525	0.35335	10.20%	0.46059	0.7672
43	γ_ϕ hole	0.525	0.08628	0.0000	0.16343	0.5279
44	no hole	0.050	0.36922	2.36%	0.24034	1.5363
44	scalar 0.525	0.525	0.21574	0.26%	0.24034	0.8977
44	γ_ϕ hole	0.525	0.07279	0.0000	0.08528	0.8535

F Exact Store-Level Deletion: Prior Work and Measured Tables

Prior Work: Uniquely represented (canonical) data structures date to Snyder [1977] and Andersson and Ottmann [1995]; *obliviousness*, a distributional condition on the memory representation which explicitly does not require canonical representation, is Micciancio’s [1997]; the weak and strong history-independence notions used here are Naor and Teague’s [2001]; the open-addressed realization is Blleloch and Golovin’s [2007], whose table is a stable matching between keys and slots under a global key priority, and it is that priority-greedy rule and that delete-time fix-up cascade we use; Blleloch et al. [2008] had already taken unique representation into computational geometry. We claim no priority over order-independent placement and no novelty for the displacement rule or its delete-time repair. New is the composition: slots that are a lattice packing, so the canonical representation carries a minimum-separation certificate; content that is a continuously decaying amplitude whose decay commutes with deletion; a canonical object that is an energy function a dynamical read relaxes into; and a price that is negative where strong history independence is known to cost as much as an exponential slowdown relative to the weak notion for comparison-based structures [Buchbinder and Petrank, 2003]. “Fix-up cascade” is our name for the NEXT-chain delete-time repair of their Figure 1; the algorithm is theirs either way.

Structural Exactness at Zero Tolerance: The architectural implementation dictates 24 exhaustive structural orders explicitly measured at $n = 4$, mapped alongside 200 independently randomized sequence orders strictly at $n \in \{8, 16, 40, 64\}$. Furthermore, we implement 100 dedicated write/delete functional interleavings and absolute mid-decay continuous deletes. Across every evaluation parameter, the resulting structural spacing enforces an exact boundary of 1.540000. On the subsequently shipped operational store architecture, the structural centers, exact payloads, physical amplitudes, and the defining active logic mask are unconditionally byte-equal, with the energy profile $V(q)$ proving purely byte-equal across 64 heavily randomized direct queries.

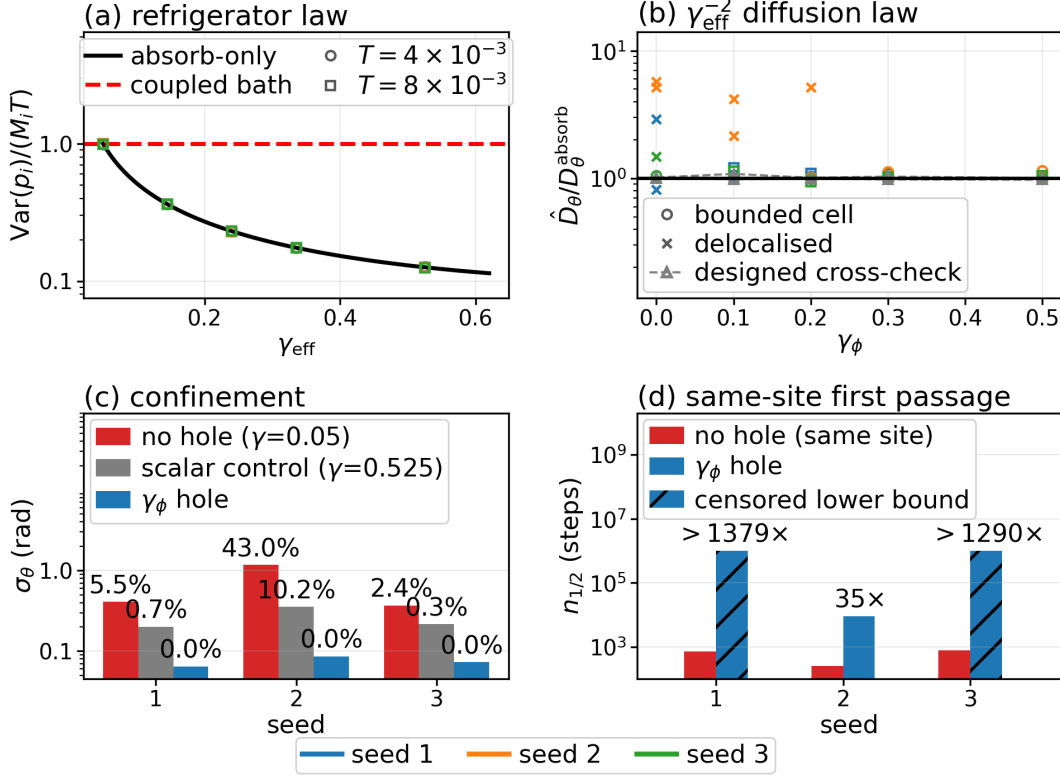


Figure 11: The vault on an emergent register; panels are labeled by the pre-registered prediction each tests. (a) The refrigerator law, $\text{Var}(p_i)/(M_i T)$ against γ_{eff} for 3 seeds $\times$ 2 temperatures, on the absorb-only curve with the coupled-bath prediction (dashed) rejected. (b) The γ_{eff}^{-2} diffusion law: bounded cells as circles and squares, delocalized cells as crosses, the designed cross-check overlaid. (c) Confinement: stationary spread and hop fraction for no hole, the scalar control and the hole. (d) Same-site first passage; hatched bars are censored lower bounds. (*Emergent MLP, 3 seeds, $T \in \{4, 8\} \times 10^{-3}$, both above T^* , $\Delta = 0.5$ rad $\Rightarrow$ evidence.*)

410 **The Negative Packing Price Constraint:** Evaluated precisely at the defined ceiling $K = 64$, the
 411 purely canonical stable-matching rule structurally admits exactly 61/64 actively offered sequential
 412 keys deterministically ($\sigma = 0$, per-offered index of 0.9531). This ratio actively climbs to absolute
 413 capacity (64/64) once physical lattice sizing is symmetrically inflated by an exactly specified
 414 coefficient of $\times 1.05$. By contrast, the stochastic refuse-and-relocate allocator architecture solely
 415 manages to admit exactly 43/64 operating on those perfectly identical geometric seeds (per-offered
 416 logic ceiling of 0.6719). It must be firmly stated that at peak physical overflow limit, operating
 417 totally independently without the waitlist active, structural exactness completely and predictably
 418 collapses: analyzing exactly 7 baseline cells receiving 8 strict capacity offers, $\text{AUC}(n_{\text{live}})$ returns
 419 1.00000 ± 0.00000 and the functional byte-equal architectural fraction plummets directly to 0.0000.

Table 7: The laundering control validation execution. Executed explicitly on the designed store featuring the shipped two-phase read geometry; evaluating 8 active targets against 3 separate seeds, mapping exactly 128 paired environments per target across 18 separate continuous amplitude limits.

Evaluated Adversary Model	Physical Decay	TTL Flag (Same Store)	Limit	Resolution Separation
White-box architecture, fully exact	1.000	1.000		0.000
Query-level membership depth, fully exact	0.983	1.000		0.017
Query-level membership depth, $\sigma_{\text{obs}} = 0.1$	0.559	0.996		0.437

Trilemma Parameters and Cost Evaluation: The physically shipped final store structure completely abandons amplitude-independent baseline address limits, generating a strict operational lifetime correlation factor computed at exactly $r = -0.85$ scaling with a_i^2 . We rigorously refute both standard previously proposed baseline repair variants: directly multiplying absolute payloads exponentially by their matched amplitudes completely corrupts the core value index criterion measured strictly at $A = 1 - \text{tol}/|a_i|$ (generating hard death physical amplitudes of exactly 0.90/0.80/0.70/0.30). *Retrieval geometry:* R_{50} contracts 1.146/1.083/0.979/0.874/0.752 as $A : 1 \rightarrow 0.5 \rightarrow 0.2 \rightarrow 0.1 \rightarrow 0.06$, a $1.52\times$ contraction from $1.146 \rightarrow 0.752$, matching the pre-registered saddle prediction at all five points (± 0.20), where a TTL vector store’s lookup radius is a constant hard step at 0.75–0.77, independent of age. Consequently, operating beneath the rigid barrier of its internal logic flag, the designated gated execution channel actively contracts the geometric value R_{50} from $1.135 \rightarrow 0.771$. Note crucially that we define no explicit deletion-cost execution statement because the core canonical synchronization mandates a complete and structural $O(n)$ rebuild evaluated entirely per operation.

G Derivations of the Closed Forms

Every closed form asserted in Secs. 2.1–2.2 and Apps. C–E follows from a single object: the damped symplectic step, linearized about the vacuum. This appendix supplies the algebra; no result here is new, and each derivation terminates in exactly the form the text prints.

Assumptions.

- *The step.* One update is: half-kick $p \mapsto p - \frac{\varepsilon}{2} \nabla V_\theta(q)$; drift $q \mapsto q' = q + \varepsilon M^{-1}p$; half-kick at q' ; damping $p \mapsto (1 - \gamma)p$, position-gated by the further factor $(1 - \gamma_\phi(q'))$ evaluated at the *updated* position when a friction field is present; and, at $T > 0$, additive noise $p \mapsto p + \sigma\xi$, $\xi \sim \mathcal{N}(0, I)$, in the Newtonian kinetic mode the Scope paragraph requires.
- *Linearization.* Retention statements expand V_θ to second order about the vacuum and diagonalize in the mass-whitened Hessian $M^{-1/2} \nabla^2 V_\theta M^{-1/2}$, whose eigenvalues are the spectral masses μ_k^2 . Write $h = \varepsilon\mu$.
- *The coset scale F .* F^2 denotes the mass-weighted squared displacement of the latent state per unit of the stored coordinate: for a stored angle on a ring of radius r_* with tied channel mass M , $F^2 = Mr_*^2$. Diffusion coefficients are per unit dynamical time $t = n\varepsilon$.

G.1 The per-mode step matrix

In a whitened normal-mode coordinate (x, p) with $V = \frac{1}{2}\mu^2 x^2$, composing the four sub-steps above gives the linear map $\begin{pmatrix} x \\ p \end{pmatrix} \mapsto J \begin{pmatrix} x \\ p \end{pmatrix}$,

$$J = \begin{pmatrix} 1 - \frac{h^2}{2} & \varepsilon \\ -(1 - \gamma)\varepsilon\mu^2(1 - \frac{h^2}{4}) & (1 - \gamma)(1 - \frac{h^2}{2}) \end{pmatrix}, \quad \text{tr } J = (2 - \gamma)\left(1 - \frac{h^2}{2}\right), \quad \det J = 1 - \gamma.$$

This is the “entirely fixed by $(\varepsilon\mu, \gamma)$ ” 2×2 matrix of the Methodological Foundations; its spectral radius sets each mode’s retention.

G.2 Conformal symplecticity

The three Verlet sub-steps are shears (each Jacobian is triangular with unit diagonal), so their composition has determinant 1. The damping sub-step is $q \mapsto q, p \mapsto (1 - \gamma)p$, contributing $(1 - \gamma)^d$; hence $\det J = (1 - \gamma)^d$. The gate multiplies each momentum component by the scalar $(1 - \gamma_\phi(q'))$; because q' is already fixed when the gate acts, its Jacobian is block-triangular and contributes exactly $(1 - \gamma_\phi(q'))^d$ — the position-gated form quoted in Sec. 2.1.

G.3 The V-curve: two branches and $\gamma_{\text{crit}} = 2\varepsilon\mu$

The eigenvalues of G.1 are $\lambda_\pm = \frac{1}{2}(\text{tr } J \pm \sqrt{\text{tr}^2 J - 4 \det J})$, and the discriminant’s sign splits the curve.

463 *Underdamped branch* ($\text{tr}^2 J < 4 \det J$): the pair is complex with $|\lambda|^2 = \det J = 1 - \gamma$, so

$$n_{1/2} = \frac{\ln 2}{-\frac{1}{2} \ln(1 - \gamma)} \xrightarrow{\gamma \ll 1} \frac{2 \ln 2}{\gamma},$$

464 independent of μ^2 and of the checkpoint — the left-branch exclusion of App. D ($\lambda_{\text{ret}} =$
 465 $\sqrt{1 - 0.002} = 0.998999499$, $n_{1/2} = 692.5$ vs. $2 \ln 2 / 0.002 = 693.1$), the dashed $2 \ln 2 / \gamma$ en-
 466 velope of Fig. 5a, and, evaluated at $\gamma = 0.05$, the mass-independent floor of 27.03 steps of App. C.

467 *Overdamped branch*: substituting $\lambda = 1 - \delta$ into $\lambda^2 - \lambda \text{tr} J + \det J = 0$ gives exactly $\delta^2 - [\gamma +$
 468 $(2 - \gamma) \frac{h^2}{2}] \delta + (2 - \gamma) \frac{h^2}{2} = 0$, whose retained (slow) root is $\delta = (2 - \gamma) h^2 / (2\gamma) (1 + O(h^2 / \gamma^2))$.
 469 Hence

$$n_{1/2} = \frac{\ln 2}{-\ln(1 - \delta)} \simeq \frac{2 \ln 2 \gamma}{(2 - \gamma) \varepsilon^2 \mu^2},$$

470 the μ^{-2} law of Sec. 2.1. Its log-slope is $d \ln n_{1/2} / d \ln \gamma = 1 + \gamma / (2 - \gamma)$: it approaches the +1
 471 asymptote only as $\gamma \rightarrow 0$ and exceeds it at every finite γ , while the underdamped slope $-\gamma / [(1 -$
 472 $\gamma)(-\ln(1 - \gamma))]$ = $-1 - \frac{\gamma}{2} + O(\gamma^2)$ hugs -1 — which is why the two measured branches are
 473 asymmetric (slopes near -1 below, $+1.1$ to $+1.3$ above, per the fit windows of Apps. C–D).

474 *The optimum*: the branches meet where $\text{tr}^2 J = 4 \det J$, i.e. $(2 - \gamma)(1 - h^2 / 2) = 2\sqrt{1 - \gamma}$, solved
 475 exactly by $h^*(\gamma) = (1 - \sqrt{1 - \gamma}) \sqrt{2 / (2 - \gamma)}$. Inverting at small h , $\gamma^* = 2\varepsilon\mu(1 + O(\varepsilon\mu))$: the
 476 retention dial's computable optimum $\gamma_{\text{crit}} = 2\varepsilon\mu$. A flat coset is the $\mu \rightarrow 0$ limit of the same curve,
 477 with $\gamma_{\text{crit}} \rightarrow 0$ — the unification claim of Sec. 2.1.

478 **G.4 The $T = 0$ latch, and why $\gamma = 0$ is not a latch**

479 At $\mu = 0$ the matrix of G.1 is triangular, $J = \begin{pmatrix} 1 & \varepsilon \\ 0 & 1 - \gamma \end{pmatrix}$, with eigenvalues 1 and $1 - \gamma$: for every
 480 $\gamma > 0$ the stored coordinate carries $|\lambda_{\text{flat}}| = 1$ *exactly* — friction cannot erase the latch (Fig. 5b),
 481 and any residual momentum dies as $(1 - \gamma)^n$. At $\gamma = 0$, however, J is a Jordan block for $\lambda = 1$:
 482 residual momentum is never removed and the coordinate drifts ballistically, $\theta_n = \theta_0 + n \varepsilon p_\theta / F^2$ —
 483 the mechanism behind the 142.7-rad $\gamma = 0$ control of App. C. With the confinement term $\alpha \|q\|^2$
 484 present, every Hessian eigenvalue is lifted by 2α , so a flat direction becomes $\mu^2 = 2\alpha$ and G.3's
 485 overdamped rate caps the lifetime at $\tau_{\text{max}} = \Gamma / 2\alpha$ (damping rate $\Gamma = \gamma / \varepsilon$).

486 **G.5 The discrete FDT scale σ^***

487 The damping-plus-noise sub-step acts on each momentum component as the linear recursion $p' =$
 488 $(1 - \gamma)p + \sigma\xi$, whose stationary variance solves $\text{Var} = (1 - \gamma)^2 \text{Var} + \sigma^2$, i.e. $\text{Var}(p) = \sigma^2 / [\gamma(2 - \gamma)]$.
 489 Matching the Gibbs (equipartition) requirement $\text{Var}(p_i) = M_i T$ of a Newtonian kinetic term forces

$$\sigma_i^* = \sqrt{M_i T \gamma (2 - \gamma)}$$

490 — the thermodynamic-consistency scale of the Scope paragraph. (For a relativistic kinetic term
 491 the stationary momentum law of this linear recursion remains Gaussian while Gibbs demands a
 492 non-Gaussian marginal, which is why $T > 0$ claims are restricted to the Newtonian mode.)

493 **G.6 The coset diffusion law and the sign flip**

494 Along a flat stored direction the force vanishes, so per step $\Delta\theta = \varepsilon p_\theta / (Mr_*)$ while p_θ follows the
 495 recursion of G.5 with autocorrelation $\rho = 1 - \gamma$ and stationary variance MT . Summing n correlated
 496 increments, $\text{Var}(\sum_{k \leq n} p_k) \rightarrow n MT \frac{1 + \rho}{1 - \rho} = n MT \frac{2 - \gamma}{\gamma}$, hence $\text{Var}(\theta_n) = \varepsilon^2 T n (2 - \gamma) / (F^2 \gamma)$,
 497 i.e. with $t = n\varepsilon$ and $\text{Var} = 2D_\theta t$:

$$D_\theta = \frac{\varepsilon T (2 - \gamma)}{2 F^2 \gamma}.$$

498 A diffusive half-life to a read tolerance Δ scales as $n_{1/2} \propto \Delta^2 / (\varepsilon D_\theta) \propto \gamma / [(2 - \gamma)T]$: increasing γ
 499 *lengthens* coset memory ($n_{1/2} \propto \gamma^{+1}$, the sign flip of Sec. 2.2), and only T erases it ($n_{1/2} \propto T^{-1}$),
 500 while the massive mode of G.3 responds with the opposite sign — the two rows of the $T > 0$ budget
 501 table.

502 G.7 The friction hole: brake, refrigerator, vault

503 Inside a uniform hole the step damps twice, $p' = (1 - \gamma_\phi)(1 - \gamma)p + \sigma^* \xi$, so the effective damping is
 504 $\gamma_{\text{eff}} = 1 - (1 - \gamma)(1 - \gamma_\phi)$ while the noise keeps the *scalar*- γ scale $\sigma^{*2} = MT\gamma(2 - \gamma)$ (absorb-only,
 505 no local re-emission). Repeating G.5 with mismatched damping and noise,

$$\text{Var}(p) = \frac{MT\gamma(2 - \gamma)}{\gamma_{\text{eff}}(2 - \gamma_{\text{eff}})} \equiv MT_{\text{local}}, \quad T_{\text{local}} = T \frac{\gamma(2 - \gamma)}{\gamma_{\text{eff}}(2 - \gamma_{\text{eff}})},$$

506 which at $\gamma = 0.05$, $\gamma_\phi \in \{0.1, 0.2, 0.3, 0.5\}$ evaluates to the absorb-prediction column
 507 0.36249/0.23082/0.17480/0.12591 of the refrigerator table, and at $T = 10^{-3}$ to $T_{\text{local}} =$
 508 1.26×10^{-4} . Repeating G.6 with $\rho = 1 - \gamma_{\text{eff}}$,

$$D_\theta = \frac{\varepsilon T \gamma(2 - \gamma)}{2F^2 \gamma_{\text{eff}}^2}, \quad \text{vault} = \frac{D_\theta(\text{no hole})}{D_\theta(\text{hole})} = \left(\frac{\gamma_{\text{eff}}}{\gamma} \right)^2$$

509 (every $(2 - \gamma)$ factor cancels), = 110.25 at $\gamma = 0.05$, $\gamma_\phi = 0.5$. The coupled-bath alternative instead
 510 rescales the noise to $\sigma^2 = MT\gamma_{\text{eff}}(2 - \gamma_{\text{eff}})$, giving $T_{\text{local}} = T$ and $D_\theta = \varepsilon T(2 - \gamma_{\text{eff}})/(2F^2 \gamma_{\text{eff}}) \propto$
 511 γ_{eff}^{-1} , a vault of $(2 - \gamma)\gamma_{\text{eff}}/[\gamma(2 - \gamma_{\text{eff}})] = 13.88$ at the same settings. The ratio of the two hypotheses
 512 is $\gamma_{\text{eff}}(2 - \gamma_{\text{eff}})/[\gamma(2 - \gamma)] = T/T_{\text{local}}$: the $7.942\times$ separation factor of App. E is the refrigerator
 513 factor, identically.

514 G.8 Decay commutes with deletion

515 Let the store state be the live key set K with per-item amplitudes $\{a_i\}$, and let the layout be
 516 the canonical placement $L(K)$ — under the stated conditions (sufficient budget, priority/attribute
 517 eviction) a function of K and the fixed geometry alone, and in particular independent of the amplitudes.
 518 Amplitude decay acts item-wise, $a_i \mapsto f(a_i)$, touching neither K nor $L(K)$; deletion removes one
 519 key and recanonicalizes, touching neither the surviving amplitudes nor their decay clocks. The two
 520 operations act on disjoint components of the state $(K, \{a_i\})$, so they commute, and the post-delete
 521 store is byte-identical to the store built from the never-inserted history with the same elapsed decay
 522 — the commutation and byte-equality invoked in Sec. 2.3 and App. F. Recency eviction makes L a
 523 function of the insertion *history* rather than of K , which is exactly why it is excluded.